Classification IV: Vertebrates

We can summarise the characteristics of the five main groups of vertebrates in a table:

Fish	Amphibians	Reptiles	Birds	Mammals
ectotherm	ectotherm	ectotherm	endotherm	endotherm
live in water	young live in water	skin covered in scales	feathers	hair or fur
have gills	adults breathe air	air breathers	air breathers	air breathers
lay multiple eggs	lay eggs without shells	lay soft-shelled eggs	lay hard-shelled eggs	live young fed on milk

1 Read the passage on fish below and give definitions of the terms in bold.

Fish are **aquatic** and are a heterogeneous group of animals – that is, not all fish fit the **criteria** used to describe fish in general. Fish have been in existence for nearly 500 million years. Their skeletons can be made of bone or **cartilage**. Most fish have a streamlined body and are propelled through the water moving their body from side to side assisted by **fins** and a tail. Fish have a heart with two chambers and obtain oxygen through **gills**. Most fish have scales (overlapping plates) covered in **mucus**. Some fish have a gas **bladder** – an organ that adjusts a fish's buoyancy to allow it to swim at different depths. Fish are ectotherms, they cannot control their internal temperature by altering their **metabolic** rate.

2 Explain the difference between ectotherms and endotherms.

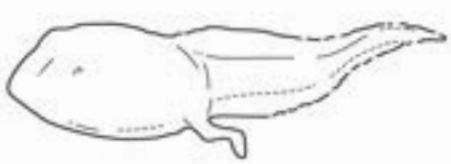
3 Name three reptiles.

Reptiles (Class *Reptilia*, from the Latin, meaning 'to crawl') are covered in scales and have a three-chambered heart. All reptiles, even those that live in water, breathe through their lungs. Most are oviparous (lay eggs) but some are viviparous (give birth to live young). Reptiles control their body temperature by outside sources such as sunlight or warm ground. They are usually found in warmer regions of the Earth. In temperate regions they tend to hibernate in colder weather, and in very hot conditions they estivate (go into a torpor). Reptiles have been on Earth for around 300 million years.

Copy the following passage and use the words listed to fill in the missing words.

air, animals, backbone, day, develop, eating, eggs, frogs, gills, moist, not, plants, shells, toads, water, heart

Amphibians are **(4)**. Examples of amphibians are **(5)** and **(6)**. Amphibians have **(7)** skin. Amphibians have a **(8)** with three chambers. Amphibians lay **(9)** in water. The eggs don't have **(10)**. The young live in **(11)** and breathe using **(12)**. Tadpoles eventually **(13)** into frogs. This change is called metamorphosis. The young tadpoles are herbivorous – they eat **(14)**. Adult amphibians have lungs and breathe **(15)**. They are usually carnivorous, **(16)** insects, worms and slugs. Amphibians do **(17)** have a steady body temperature.
The salamander is an amphibian. Amphibians are vertebrates, meaning they have a **(18)**.
They tend to be more active at night, rather than during the **(19)**.



20 Name three amphibians not already mentioned.

21 Would you expect to find reptiles living in or near the Arctic or Antarctic? Explain your answer.

22 Describe what is meant by *hibernate* and *estivate*.

23 Suggest an advantage of hibernation and estivation.

24 Suggest why reptiles have not really changed over time.

© Wendy St Germain, Peter Gribben and Learning Essentials 2003.

© Blake Education ISBN 978 1 92561 063 5 Assignments in Biology Book 1